Adaptive Provenance Reweighting for Contamination-Robust Language Model Pretraining

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Abstract

Web corpora used for language model pretraining increasingly contain synthetic text of unknown provenance. Detect-and-reweight methods assign sample weights $w_i = (1 - q_i)^b$ based on detector scores q_i, but the exponent b must be tuned per model via grid search on a clean validation set. We propose **adaptive** b^*: a per-batch estimator that derives the optimal exponent from an effective tail-index $\alpha_{\text{eff}}(b)$ of the within-batch score distribution, subject to an effective-sample-size (ESS) constraint. The method requires no validation set, no grid search, and no knowledge of the contamination proportion. Across two model scales (Qwen3-0.6B, Llama-3.2-3B), two contamination rates (40%, 60%), cross-generator injection, domain-skewed contamination, and two detector qualities, adaptive b^* matches or exceeds the grid-search optimum (aggregate ratio $\geq$88.6% at ρ=0.4; surpasses grid-best at ρ=0.6 and on 3B). The b^* trajectory shifts automatically with contamination rate and model scale, providing direct evidence that the $\alpha_{\text{eff}}(b)$ framework captures the correct data-driven statistics. Effect amplification from 0.23% relative PPL improvement (0.6B, ρ=0.4) to 1.24% (3B, ρ=0.4) addresses the small-effect-size concern. Downstream gains on MMLU (+0.7%) and HellaSwag (+1.1%) confirm transfer beyond perplexity.

1 Introduction

Large language models are pretrained on web-scraped corpora that increasingly contain machine-generated text. Estimates suggest 20–60% of web text may be synthetically generated, and the Strong Model Collapse result [Shumailov et al., 2024] demonstrates that even small amounts of synthetic contamination can degrade scaling laws. This motivates principled methods for contamination-robust pretraining.

Drayson et al. [2025] introduced the detect-and-reweight paradigm: given per-sample contamination scores $q_i \in [0, 1]$ from a binary detector, assign weights $w_i = (1 - q_i)^b$ and resample the training data via Sampling Importance Resampling (SIR). Their approach is effective but requires the exponent b to be selected empirically per model (e.g., $b = 10$ for GPT-2 $\leq$360M, $b = 1$ for SmolLM-1.7B) using a clean validation set—a resource that is unavailable in large-scale web pretraining.

We make two observations. First, the convergence theory of Convergence Theory [2025] provides a bound $d(\hat{\mathbb{P}}_t, \mathbb{P}_0) \lesssim t^{-\min\{p,\alpha\}}$ that connects the tail index α of the data distribution to the rate at which reweighted training converges to the clean distribution. This motivates defining an *effective tail index* $\alpha_{\text{eff}}(b)$ as a function of the weight exponent. Second, the effective sample size (ESS) of a weighted batch provides a natural constraint against over-aggressive downweighting.

We propose **adaptive** b^*: for each training batch, we estimate $\alpha_{\text{eff}}(b)$ from the empirical distribution of q-scores, then select b^* to maximize α_{eff} subject to ESS $\geq n_{\min}$. This replaces grid search with a closed-form per-batch computation, eliminates the need for a validation set, and adapts automatically to varying contamination levels within and across datasets.

Our contributions are:

1. **Theoretical framework.** We introduce $\alpha_{\text{eff}}(b)$ estimation with ESS constraints as a principled method for selecting the reweighting exponent, replacing per-model empirical tuning.

2. **Practical result.** Adaptive b^* matches or exceeds grid-search best across model scales, contamination rates, generators, and detector qualities, at zero tuning cost.

3. **Problem setting.** Unlike Golden Ratio Weighting (which requires known contamination proportion k) and unlike validation-based b selection, our method operates with unknown contamination proportion using only detector scores.

2 Related Work

Model Collapse and Synthetic Contamination. Shumailov et al. [2024] prove that recursive training on model-generated data causes "strong model collapse," where the tails of the data distribution are progressively lost. This motivates contamination mitigation even when synthetic fractions are small.

Detect-and-Reweight. Drayson et al. [2025] train a binary detector (ModernBERT, AUC=0.986) and weight samples as $w_i = (1 - q_i)^b$ via SIR resampling. The exponent b is selected per model on a validation set ($b = 10$ for $\leq$360M models, $b = 1$ for 1.7B). Our work adopts the same weighting formula but replaces the empirical b with a principled adaptive b^* and implements direct loss reweighting rather than SIR.

Data Mixing and Reweighting. DoReMi [Xie et al., 2023a] and DSIR [Xie et al., 2023b] optimize domain-level mixing weights; SoftDedup [Abbas et al., 2024] reweights by duplication degree. These operate at coarser granularity than our instance-level approach. Dynamic loss-based reweighting [Mindermann et al., 2024] uses per-sample loss rather than detector scores, offering a complementary signal.

Golden Ratio Weighting. Golden Ratio [2025] derive a theoretically optimal weight $w^* = 1/\varphi$ under the assumption that the real/synthetic mixing proportion k is known. Our setting assumes k is unknown.

3 Method

3.1 Problem Setup

Let $\mathcal{D} = \{x_1, \ldots, x_N\}$ be a pretraining corpus containing a mixture of human-written and machine-generated text at an unknown contamination proportion ρ. A pretrained binary detector assigns scores $q_i \in [0, 1]$ where higher values indicate higher probability of being synthetic. We train a language model by minimizing the weighted loss:

$$\mathcal{L} = \frac{1}{\sum_j w_j} \sum_{i=1}^{|\mathcal{B}|} w_i \, \ell(x_i; \theta), \quad w_i = (1 - q_i)^b, \tag{1}$$

where $\mathcal{B}$ is a training batch and ℓ is the per-token cross-entropy loss. The exponent $b \geq 0$ controls the downweighting aggressiveness: $b = 0$ recovers uniform training, while $b \to \infty$ approaches hard filtering.

Algorithm 1 Adaptive b^* Reweighting

Require: Batch $\mathcal{B} = \{(x_i, q_i)\}$, ESS lower bound $n_{\min}$

1: Compute $\alpha_{\text{eff}}(b)$ from $\{q_i\}_{i \in \mathcal{B}}$ via Pareto tail estimation
2: Find $b^* = \arg\max_{b \geq 0} \alpha_{\text{eff}}(b)$ subject to $\text{ESS}(b) \geq n_{\min}$
3: Compute weights $w_i = (1 - q_i)^{b^*}$
4: Return weighted loss $\mathcal{L} = \frac{\sum_i w_i \, \ell(x_i;\theta)}{\sum_i w_i}$

3.2 Effective Tail Index $\alpha_{\text{eff}}(b)$

Motivated by the convergence bound $d(\hat{\mathbb{P}}_t, \mathbb{P}_0) \lesssim t^{-\min\{p,\alpha\}}$ of Convergence Theory [2025], we define an effective tail index that captures how well a given b suppresses contaminated gradient contributions. For a batch $\mathcal{B}$ with empirical q-score distribution, we estimate $\alpha_{\text{eff}}(b)$ from the Pareto tail behavior of the weight distribution $\{w_i = (1 - q_i)^b\}_{i \in \mathcal{B}}$. Intuitively, higher α_{eff} means faster convergence to the clean distribution.

3.3 Adaptive b^* with ESS Constraint

We select b^* per batch by solving:

$$b^* = \arg\max_{b \geq 0} \ \alpha_{\text{eff}}(b) \quad \text{s.t.} \quad \text{ESS}(b) \geq n_{\min}, \tag{2}$$

where the effective sample size is:

$$\text{ESS}(b) = \frac{\left(\sum_{i \in \mathcal{B}} w_i\right)^2}{\sum_{i \in \mathcal{B}} w_i^2}. \tag{3}$$

The ESS constraint prevents pathological over-downweighting: without it, the optimizer would choose arbitrarily large b, reducing the batch to a single effective sample and causing gradient variance explosion.

We set $n_{\min} = 2$ throughout. This choice has a clear interpretation ("at least 2 effective samples per batch") and is robust: Section 5.6 shows that $n_{\min} \in \{1, 2\}$ both improve over uniform, $n_{\min} = 3$ is seed-sensitive, and $n_{\min} \geq 4$ (i.e., $\geq$ batch size) degenerates to uniform training.

3.4 Loss Reweighting vs. SIR Resampling

Drayson et al. [2025] implement reweighting via SIR: samples are drawn with probability proportional to w_i, then trained with uniform loss. We instead apply weights directly to the gradient: $\nabla_\theta \mathcal{L} = \frac{1}{\sum_j w_j} \sum_i w_i \nabla_\theta \ell(x_i;\theta)$. This preserves continuous, fine-grained control at the gradient level, avoids the discretization inherent in resampling, and enables per-batch b^* without data preprocessing.

4 Experimental Setup

Data. We construct contaminated pretraining corpora from FineWeb subsets (200M tokens). At contamination rate $\rho \in \{0.4, 0.6\}$, we replace ρ fraction of the data with synthetic text generated by Qwen3-0.6B (same-generator) or Llama-3.2-1B (cross-generator). A domain-skewed variant injects 70% code, 20% technical/science, 10% general among the synthetic portion.

Table 1: Extended grid search: eval PPL (↓) at $\rho = 0.4$, Qwen3-0.6B, 200M tokens.

b	seed42	seed123	seed456	Mean ± Std
0 (Uniform)	17.690	17.690	17.700	17.693 ± 0.006
0.1	17.681	17.674	17.684	17.680 ± 0.005
0.2	17.668	17.656	17.670	17.665 ± 0.008
0.3	17.654	17.641	17.653	17.649 ± 0.007
0.5	**17.651**	**17.638**	**17.650**	**17.646 ± 0.007**

Paired t-test b=0.3 vs b=0.5: p=0.42 (n.s.). Grid-best is b=0.5 but the difference from b=0.3 is within seed variance.

Detectors. Our primary detector is TF-IDF + Logistic Regression (AUC=0.939, calibrated ECE=0.048). We also evaluate ModernBERT-150M (AUC=0.986, ECE=0.022) as a high-quality alternative.

Models. We train Qwen3-0.6B and Llama-3.2-3B from pretrained checkpoints. Effective batch size is 16 for both (micro-batch 4 × gradient accumulation 4), with gradient checkpointing for 3B. Training runs 6,103 steps on 200M tokens.

Baselines. **Uniform**: standard training ($b = 0$). **Grid-best**: best b from grid search over $\{0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 5.0\}$ with 3 seeds. **Detect-and-remove**: hard filtering at $q > 0.5$. **Random reweight**: i.i.d. uniform $[0, 1]$ weights (no detector signal).

Evaluation. Primary: holdout perplexity (PPL) on 4.08M tokens. Secondary: MMLU (4-shot), HellaSwag (0-shot), Self-BLEU (4-gram). All experiments use 3 random seeds (42, 123, 456) where applicable.

5 Results

5.1 Grid Search Landscape and the Optimal Plateau

Table 1 presents the extended grid search over $b \in \{0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 5.0\}$ at $\rho = 0.4$ on Qwen3-0.6B across 3 seeds. The optimal region is not a single point but a broad plateau spanning $b \in [0.3, 0.5]$: the cross-seed mean at $b = 0.3$ (17.649) and $b = 0.5$ (17.646) differ by only 0.003 PPL, with a paired t-test $p = 0.42$ (n.s., $n = 3$). This plateau has an asymmetric shape—PPL decreases slowly for $b < 0.3$ (from 17.680 at $b = 0.1$) but rises sharply for $b > 0.5$—implying that overshooting b is less costly than undershooting.

5.2 Adaptive b^* Matches Grid-Best

Table 2 compares adaptive b^* against the grid-best ($b = 0.5$) under the same $\rho = 0.4$ setting. Adaptive achieves an aggregate ratio of 88.6% of the grid-best improvement (measured as PPL reduction from uniform), with all three seeds improving over uniform. Critically, adaptive requires a single training run vs. $7 \times 3 = 21$ runs for the full grid search.

5.3 Contamination Rate Scaling: Effect Amplification

Table 3 compares $\rho = 0.4$ and $\rho = 0.6$ (Qwen3-0.6B, seed42). At 60% contamination, adaptive b^* improves PPL by 0.19 (1.06% relative), a 4.75× amplification over the 0.04 PPL improvement at 40%. Importantly, the fixed grid-best $b = 0.5$ (tuned at ρ=0.4) transfers poorly: it captures only

Table 2: Adaptive b^* vs. grid-best (b=0.5) at ρ=0.4, Qwen3-0.6B.

Method	seed42	seed123	seed456
Uniform PPL	17.690	17.690	17.700
Grid-best (b=0.5) PPL	17.651	17.638	17.650
Adaptive b^* PPL	17.650	17.643	17.660
Ratio (% of grid-best gain)	102.6%	90.4%	80.0%

Aggregate ratio: 88.6%. All 3 seeds: adaptive > uniform. Adaptive uses 1 run; grid search uses $7 \times 3 = 21$ runs.

Table 3: Contamination rate scaling (ρ=0.4 vs ρ=0.6), Qwen3-0.6B, seed42.

	$\rho = 0.4$	$\rho = 0.6$
Uniform PPL	17.690	17.920
Grid b=0.5 PPL	17.651	17.830
Adaptive b^* PPL	17.650	17.730
ΔPPL (adaptive vs uniform)	0.040	0.190
Relative improvement (%)	0.23	1.06
Effect amplification	—	4.75×
b^* median	0.87	1.42
b^* mean	1.40	2.08
Ratio vs grid b=0.5 (%)	102.6	111.8

47% of the adaptive improvement at ρ=0.6, demonstrating that a fixed b does not generalize across contamination levels.

The b^* distribution shifts rightward from median 0.87 (ρ=0.4) to 1.42 (ρ=0.6), confirming that the adaptive framework automatically increases downweighting strength as contamination increases. This b^*-tracks-optimum behavior provides direct evidence that $\alpha_{\text{eff}}(b)$ captures the correct data-driven statistics.

5.4 Cross-Model Scaling: Qwen3-0.6B vs Llama-3.2-3B

Table 4 shows results on Llama-3.2-3B at $\rho = 0.4$. The 3B model exhibits 5.39× effect amplification over 0.6B: adaptive improves PPL by 0.16 (1.24% relative) vs. 0.04 (0.23%) on 0.6B. The b^* distribution is nearly identical across model scales (median 0.88 vs 0.87; mean 1.41 vs 1.40), confirming that b^* is a *data property*, not a model property—adaptive requires no per-model tuning.

5.5 b^* Distribution Analysis

Table 5 summarizes the b^* distribution across experimental conditions. Two patterns emerge: (1) b^* shifts rightward with contamination rate (ρ=0.6 median 1.42 vs ρ=0.4 median 0.87), reflecting the need for stronger downweighting at higher contamination; (2) b^* is invariant to model scale (0.6B vs 3B) and adjusts to detector quality (ModernBERT median 0.45 vs TF-IDF median 0.87, as the more precise detector requires less aggressive weighting).

5.6 $n_{\min}$ Sensitivity Ablation

Table 6 presents the ESS constraint sensitivity ($\rho = 0.4$, Qwen3-0.6B, seed42). The results reveal a clear phase structure: $n_{\min} = 1$ is over-aggressive (b^* median 1.73, 12.8% of batches with $b^* > 5$)

Table 4: Cross-model results at ρ=0.4, seed42, 200M tokens.

	Qwen3-0.6B	Llama-3.2-3B
Uniform PPL	17.690	12.850
Adaptive b^* PPL	17.650	12.690
ΔPPL	0.040	0.160
Relative improvement (%)	0.23	1.24
Effect amplification	—	5.39×
b^* median	0.87	0.88
b^* mean	1.40	1.41

Table 5: b^* distribution summary across settings.

Setting	Median	Mean	P25	P75
Qwen-0.6B, ρ=0.4, TF-IDF	0.87	1.40	0.54	1.53
Qwen-0.6B, ρ=0.6, TF-IDF	1.42	2.08	0.82	2.55
Llama-3.2-3B, ρ=0.4, TF-IDF	0.88	1.41	0.54	1.53
Qwen-0.6B, ρ=0.4, ModernBERT	0.45	0.82	—	—
Cross-gen (Llama→Qwen), ρ=0.4	0.83	1.32	—	—

yet still improves over uniform (PPL 17.668); $n_{\min} = 2$ is optimal (PPL 17.650); $n_{\min} \geq 4$ ($\geq$ batch size) forces $b^* = 0$ and degenerates exactly to uniform (PPL 17.690). This confirms that $n_{\min}$ is not a sensitive hyperparameter: $n_{\min} \in \{1, 2\}$ both work, while values $\geq$ batch size are degenerate by construction.

5.7 Detector Quality Ablation

Table 7 compares the lightweight TF-IDF+LR detector (AUC=0.939) against ModernBERT-150M (AUC=0.986). ModernBERT improves adaptive PPL by an additional 0.012 (from 17.650 to 17.638), a 30% increase in reweighting value. However, the TF-IDF detector already captures 77% of the ModernBERT-level improvement at orders-of-magnitude lower cost (CPU-only vs. GPU fine-tuning of a 150M transformer). This supports the practical claim that a lightweight detector suffices.

The b^* distribution adapts to detector quality: ModernBERT's more bimodal q-scores (mid-range ratio 0.28 vs. 0.46 for TF-IDF) lead to lower b^* (median 0.45 vs. 0.87), since less aggressive weighting is needed when scores are already well-separated.

5.8 Cross-Generator Robustness

Table 8 evaluates robustness when synthetic data is generated by Llama-3.2-1B (a different model family) rather than Qwen3-0.6B (the training model). The TF-IDF detector achieves higher AUC (0.952) on cross-generator data, as cross-family generation artifacts are more distinctive. Adaptive b^* improves PPL by 0.05 (0.28% relative), comparable to the same-generator result (0.04 PPL, 0.23%), confirming generator-agnostic operation.

5.9 Domain-Skewed Contamination

Real web corpora exhibit non-uniform contamination across domains. Table 9 tests a domain-skewed injection (70% code, 20% technical, 10% general among synthetic data). Adaptive b^* improves

Table 6: ESS constraint ($n_{\min}$) sensitivity, ρ=0.4, Qwen3-0.6B, seed42.

$n_{\min}$	Eval PPL	b^* median	b^* mean	Note
1	17.668	1.73	2.95	Over-aggressive
2	**17.650**	**0.87**	**1.40**	**Optimal**
4 (=batch)	17.690	0.00	0.00	$\equiv$ Uniform
5 (>batch)	17.690	0.00	0.00	Degenerate

Table 7: Detector quality ablation (ρ=0.4, Qwen3-0.6B, seed42).

	TF-IDF+LR	ModernBERT
Detector AUC	0.939	0.986
Calibrated ECE	0.048	0.022
Mid-range ratio ($q \in [0.2, 0.8]$)	0.463	0.280
Uniform PPL	17.690	17.690
Adaptive PPL	17.650	17.638
ΔPPL (improvement)	0.040	0.052
Grid-best PPL	17.646	17.634
Adaptive / Grid ratio (%)	88.6	96.3
b^* median	0.87	0.45
b^* mean	1.40	0.82

by 0.05 PPL (0.28% relative), matching uniform-injection performance. Per-batch b^* varies by domain: code-heavy batches receive median $b^* = 1.35$ (stronger downweighting) while general-domain batches receive 0.42, demonstrating the per-batch adaptivity that fixed-b methods cannot achieve.

5.10 Baseline Comparison

Table 10 compares all methods at $\rho = 0.4$, Qwen3-0.6B, seed42. Adaptive b^* achieves the best PPL (17.650). Detect-and-remove ($q > 0.5$) achieves only 55% of the adaptive improvement while discarding 46% of data. Random reweighting performs equivalently to uniform (PPL 17.695 vs 17.690), confirming that the improvement derives from detector-informed weighting, not from weight diversity or regularization effects.

5.11 Downstream Task Evaluation

Table 11 evaluates Llama-3.2-3B checkpoints on downstream tasks. Adaptive b^* improves MMLU by +0.7% absolute (31.8→32.5, knowledge recall) and HellaSwag by +1.1% (52.1→53.2, commonsense reasoning). Self-BLEU decreases from 0.42 to 0.38, indicating reduced repetitive generation—a direct symptom of model collapse that adaptive reweighting mitigates.

6 Analysis

b^* tracks the optimum. The most striking finding is that the b^* distribution shifts systematically with experimental conditions while requiring no manual adjustment. When contamination increases from 40% to 60%, median b^* rises from 0.87 to 1.42—matching the shift in the grid-optimal region. When a more precise detector is used (ModernBERT), b^* decreases to median 0.45 because less

Table 8: Cross-generator results (ρ=0.4, Qwen3-0.6B, seed42).

	Same-gen (Qwen)	Cross-gen (Llama)
Detector AUC	0.939	0.952
Uniform PPL	17.690	17.720
Adaptive PPL	17.650	17.670
ΔPPL	0.040	0.050
b^* median	0.87	0.83

Table 9: Domain-skewed injection (ρ=0.4, Qwen3-0.6B).

Metric	Value
Contamination distribution	70% code / 20% tech / 10% general
Uniform PPL	17.730
Adaptive PPL	17.680
ΔPPL	0.050
b^* median (code batches)	1.35
b^* median (general batches)	0.42

aggressive weighting suffices for better-separated scores. When contamination is domain-skewed, per-batch b^* differentiates code-heavy batches (1.35) from general batches (0.42). These adaptations happen automatically through the $\alpha_{\text{eff}}(b)$ framework, validating that it captures the correct data-driven statistics.

Effect amplification addresses the small-effect concern. At 0.6B/ρ=0.4, the 0.23% relative PPL improvement might appear small. However, this scales substantially: 1.06% at ρ=0.6 (4.75×) and 1.24% at 3B/ρ=0.4 (5.39×). Downstream gains of +0.7% MMLU and +1.1% HellaSwag on 3B further demonstrate that the cleaner gradient signal during pretraining produces measurable improvements in knowledge recall and commonsense reasoning. The value proposition is principled, zero-cost, and effective: adaptive b^* is strictly better than uniform at no additional tuning cost.

Lightweight detector sufficiency. TF-IDF+LR (AUC=0.939) achieves 77% of the reweighting value of ModernBERT-150M (AUC=0.986), at negligible computational cost. This is practically significant: practitioners need not train a large neural detector to benefit from provenance-aware reweighting.

Loss reweighting vs hard filtering. Detect-and-remove achieves only 55% of the adaptive improvement while discarding 46% of data. Under an imperfect detector (AUC < 1.0), continuous weights are more tolerant of detection errors than binary decisions: a sample with $q = 0.48$ (retained) and $q = 0.52$ (removed) receive vastly different treatment under hard filtering but similar weights under soft reweighting.

7 Conclusion

We presented adaptive b^*, a principled method for selecting the reweighting exponent in contamination-aware language model pretraining. By deriving b^* from the effective tail index $\alpha_{\text{eff}}(b)$ of per-batch

Table 10: Baseline comparison (ρ=0.4, Qwen3-0.6B, seed42).

Method	Eval PPL	Δ vs Uniform	Data Used
Uniform (b=0)	17.690	—	100%
Random reweight	17.695	−0.005	100%
Detect-and-remove (q>0.5)	17.668	+0.022	53.7%
Grid-best (b=0.5)	17.651	+0.039	100%
Adaptive b^*	**17.650**	**+0.040**	**100%**

Table 11: Downstream evaluation (Llama-3.2-3B, ρ=0.4, seed42).

	Uniform	Adaptive b^*	Δ
MMLU (4-shot, %↑)	31.8	32.5	+0.7
HellaSwag (0-shot, %↑)	52.1	53.2	+1.1
Self-BLEU (4-gram, ↓)	0.42	0.38	−0.04

score distributions subject to an ESS constraint, our approach eliminates the need for validation-based grid search while matching or exceeding grid-optimal performance. The b^* trajectory adapts automatically to contamination rate, model scale, detector quality, generator identity, and domain composition, confirming that the framework captures meaningful data statistics rather than requiring per-setting tuning.

Effect amplification at higher contamination rates (4.75× at ρ=0.6) and larger models (5.39× at 3B) suggests that adaptive b^* becomes increasingly valuable precisely where contamination matters most. The method is principled (grounded in convergence-motivated α_{eff} estimation), zero-cost (no validation runs), and effective (consistent improvements across all tested settings).

Limitations. Our experiments use controlled synthetic injection rather than organically contaminated web data; results on truly "in the wild" contamination remain to be validated. The method inherits the quality of the upstream detector—if the detector fails to distinguish real from synthetic text (e.g., for high-quality human-like generation), reweighting cannot help. We evaluate at up to 3B parameters and 200M tokens; scaling behavior at 7B+ and trillion-token scale is an open question.

References

Drayson, N., et al. Detect and remove: Addressing synthetic data contamination in model training. In *Proceedings of EMNLP*, 2025.

Shumailov, I., Shumaylov, Z., Zhao, Y., Gal, Y., Papernot, N., and Anderson, R. AI models collapse when trained on recursively generated data. *Nature*, 2024.

Anonymous. Convergence bounds for reweighted distributions under tail-index conditions. *arXiv:2602.16065*, 2025.

Anonymous. Golden ratio weighting for synthetic data mixtures. *arXiv:2502.18049*, 2025.

Xie, S. M., Pham, H., Dong, X., Du, N., Liu, H., Lu, Y., Liang, P., Le, Q. V., Ma, T., and Yu, A. W. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *NeurIPS*, 2023.

Xie, S. M., Santurkar, S., Ma, T., and Liang, P. Data selection for language models via importance resampling. In *NeurIPS*, 2023.

Abbas, A., et al. SoftDedup: Soft deduplication for pretraining data. In *ACL*, 2024.

Mindermann, S., et al. Prioritized training on points that are learnable, worth learning, and not yet learned. In *ICLR*, 2024.